

Project Design Phase-II

Data Flow Diagram & User Stories

Project Title	Script-Controlled ACL – Restrict Record Access Based on Field Value
Team ID	SWTID-2026-1695
Project Name	Skill Wallet

Data Flow Diagram

The Data Flow Diagram represents how a user's request to access a Skill Wallet Request record flows through the ServiceNow ACL engine before the record is returned or blocked.

- User (Admin / Department User) initiates a Read, Create, Write, or Delete action on the Skill Wallet Request table
- ServiceNow Platform intercepts the action and triggers the matching operation-specific Script-Controlled ACL
- ACL Script checks: (a) does the user have the u_swt_admin role? (b) if not, does the record's Department field value match the user's own department?
- Decision: Access Granted → record is returned/created/updated/deleted; Access Denied → ServiceNow blocks the action and returns a security error

[SCREENSHOT PLACEHOLDER: Data Flow Diagram – User → ServiceNow ACL Engine → Skill Wallet Request Table → Access Granted/Denied]

User Stories

User Type	Epic	Story #	User Story / Task	Acceptance Criteria	Priority	Release
System Admin	FR-1	USN-1	As a System Admin, I can create user records for different departments so that access control can be tested.	Users created in sys_user with correct Department value	High	Sprint-1
System Admin	FR-1	USN-2	As a System Admin, I can create custom roles (u_swt_admin, u_swt_user) so that I can assign appropriate access levels.	Roles created and assignable to users	High	Sprint-1
System Admin	FR-2	USN-3	As a System Admin, I can create the Skill Wallet Request table with required fields so that requests can be stored and access-restricted.	Table u_skill_wallet_request created with Employee, Department, Skill Name, Wallet Points, Status fields	High	Sprint-1
Department User / Admin	FR-3	USN-4	As a user, I can only read Skill Wallet Request records that belong to my own department, unless I am an admin.	Read blocked for non-matching department; allowed for matching department or admin	High	Sprint-2
Department User / Admin	FR-4	USN-5	As a user, I can only create a Skill Wallet Request record for my own department, unless I am an admin.	Create blocked for mismatched department; allowed for matching department or admin	High	Sprint-3
Department User / Admin	FR-5	USN-6	As a user, I can only update Skill Wallet Request records belonging to my own department, unless I am an admin.	Write blocked for non-matching department; allowed for matching department or admin	High	Sprint-3
Admin only	FR-6	USN-7	As an Admin, I am the only one who can delete a Skill Wallet Request record.	Delete blocked for all non-admin users regardless of department	Medium	Sprint-4

User Type	Epic	Story #	User Story / Task	Acceptance Criteria	Priority	Release
Project Team	Milestone-7	USN-8	As the Project Team, I can validate and document the complete solution so the outcome demonstrates correct field-value-based access control.	All test cases pass; documentation completed	Medium	Sprint-4